

Northeastern University, Department of Mathematics

MATH 5110: Applied Linear Algebra and Matrix Analysis. (Summer A 2026)

Instructor: He Wang

Email: he.wang@northeastern.edu

Course Webpage: on Canvas.

Office: NI 550

Office Hours: See Canvas

Recommended Texts and other Sources:

- *Finite-dimensional linear algebra*, Mark S. Gockenbach, CRC Press. (Primary source.)
- *Introduction to Linear Algebra*, Gilbert Strang, Wellesley-Cambridge Press
- *Applied Linear Algebra and Matrix Analysis*, Shores, Thomas S., Springer
- *Applied Linear Algebra*, Olver, Peter J., Shakiban, Chehrzad, Springer
- *A Second Course in Linear Algebra*, S. R. Garcia, R. A. Horn, Cambridge University Press
- *Matrix Analysis and Applied Linear Algebra*, C. D. Meyer, SIAM, 2000.
- Advanced Linear Algebra by Yisong Yang (2ed edition)
- Supplementary *lecture notes* will be available on the course webpage.
- other textbooks: see my recommended reading list:
https://docs.google.com/spreadsheets/d/1j8LcyTxBTZ5Nh0-P1rb6NT2x_2iXFp1j8ptpvHYQWGA/edit?usp=sharing

We use Gockenbach's book as the primary reference, however, text books are not required. Most of the Ebooks can be downloaded for free from Northeastern Library website. You can buy any (soft cover/student copy) printed book published by Springer from the link in Northeastern Library within \$25 if the library bought the ebook.

This syllabus is subject to reasonable changes at the discretion of the instructor. (Check Canvas for the updated version.)

Recommended background: Undergraduate-level linear algebra. E.g., Math 2331 with textbook *Linear algebra with applications*, O. Bretscher. Some basic knowledge of understanding and writing proofs.

Overview: This course provides a rigorous treatment of the concepts and computational techniques of linear algebra. The major goal is to master the fundamental theory and techniques that underlie the applications of linear algebra in computer science, statistics, engineering, etc. Another goal is to provide a solid foundation for studying other subjects later. The course is designed for graduate students in Applied Mathematics, Computer Science, and students who wish to obtain deeper insights beyond an undergraduate level linear algebra.

List of main topics:

1. **Linear system, matrix algebra:** Linear systems and matrix equations; row reduced echelon form; existence and uniqueness of solutions; matrix operations including sum, scalar product, matrix product, inverse, transpose.
2. **Linear spaces over a field:** linear vector spaces; subspaces; spanning sets, independence, nullspace of matrix; basis and dimension; coordinate; rank-nullity theorem; linear transformations; change of basis; chain complex and homology.
3. **Jordan normal forms:** determinant; eigenvalues and eigenvectors; eigenbasis; eigenspaces; algebraic and geometric multiplicity; solution of iterated linear system; singularity, invertibility, rank etc; similarity; diagonalization; powers of a matrix; Jordan normal form and the Cayley Hamilton Theorem.

Applications to Discrete dynamical systems: positive systems; Perron-Frobenius Theorem; stochastic matrix; Markov chain.
4. **Inner product spaces:** Inner product, norm, Cauchy-Schwarz inequality; orthogonal projection onto subspace; orthogonal basis; Gram-Schmidt process; symmetric matrix, diagonalization; positive definite; variational formulas for eigenvalues, Singular value decomposition and data compression.

Applications to principal component analysis (PCA) for data analysis mean, optimal orthogonal projections, principal components, PCA in Statistics, covariance matrix, optimal linear combinations of random variables, principal components, interpretation, Fast Fourier Transform.
5. **Matrix Factorization** including Jordan, Schur, LU, QR, SVD, etc.
6. **Extra topics as time allows** (Also suggested for final projects): Hilbert Spaces, Calculus for matrix-valued functions; Neural networks and back propagation equation; Hilbert Spaces; Convergence of sequences and series in a normed vector space; Linear Programming; application to spectral graph theory; Wavelet Decomposition, Haar Bases and Haar wavelets; Hadamard matrices; The group of unit quaternions, $SU(2)$, and the representation of rotations in $SO(3)$ by unit quaternions; The geometry of the orthogonal groups $O(n)$ and $SO(n)$, and of the unitary groups $U(n)$ and $SU(n)$; Exterior algebra and tensor algebra; Multilinear Algebra; Affine space and affine maps; Duality norm and dual norms, etc.

Software: MATLAB or/and Python will be used throughout the course. Students should be prepared to use *Matlab/Python* for assignments and for projects.

Northeastern University has an active MATLAB and Simulink Campus-Wide License. Go to <https://www.mathworks.com/products/matlab.html> and use your Northeastern email address to create an account. You should be able get a free license for the Matlab product. Download and install Matlab on your computer.

Grade breakdown:

1. Attendance and Class discussion participation (5%)
2. Homework (25%): There will be 6-7 homework assignments which will focus on the theory. Homework will be assigned each week or two. You have one week to finish the homework. In addition to being mathematically correct, your write-ups are expected to show a high level of clarity and completeness.

3. Computer Labs (15%): There will be roughly 4-5 labs will focus on the applications to real world problems using Matlab/Python. Computer labs are designed to help you learn the concepts and techniques of linear algebra by using them in an computer environment and master the computational tools for important applications of linear algebra. Labs will be graded based on completeness, correctness and clarity in written.
4. Tests/Quizzes (35%): There will be 2-4 in class tests/quizzes.
5. Final project (20%). Project can be done in a group within 1-4 students. A final project can be a computational analysis of a data set using sufficiently complicated or novel techniques from this course. It can also be a theoretical presentation of a topic not covered in this course with a case study.

To receive full credit, homework/lab assignments must normally be handed in when due. Assignments may be turned in late with a valid excuse; this should be discussed with me and approved in advance if possible or otherwise at the earliest opportunity.

Exam/Tests makeup policy: Makeup exams will be allowed only in the event of a documented medical or other unforeseen emergency. You are responsible for avoiding foreseeable conflicts with the exams.

Collaboration:

You are welcome, even encouraged, to collaborate on the homework and lab assignments, though we urge you to first attempt working out all of the problems by yourself. However, you are expected to write answers yourself and understand everything that you hand in. Copy results from any sources will be considered as violating Academic Integrity. Collaboration is not allowed on the quizzes and exams.

University Academic Integrity Policy: The university's academic integrity policy at OS-CCR (<http://www.northeastern.edu/osccr/academic-integrity-policy>) discusses actions regarded as violations and their consequences for students.

Title IX: The University strictly prohibits sex or gender discrimination in all university programs and activities. Information on how to report an incident of such discrimination (which includes sexual harassment and sexual assault) is located at <http://www.northeastern.edu/titleix>.

Students with Disabilities: Students who have disabilities who wish to receive academic services and accommodations should follow the standard Disabilities Resource Center (DRC) procedures, <http://www.northeastern.edu/drc/getting-started-with-the-drc>.

We Care: Help with academics, adjusting to campus life, social life, missing or stolen possessions, travel issues, financial issues, or any other concerns, **We Care:** <https://we-care.studentlife.northeastern.edu/>

College of Science Policies: The current College of Science Academic Course Policies are available at

<https://cos.northeastern.edu/wp-content/uploads/2012/10/COS-teaching-policies-April-2017.pdf>

TRACE: Every student is expected to complete the online TRACE survey at the end of the semester.

A tentative schedule of topics and tasks:

(Will be updated as course proceeds. Check Canvas for updated schedule)

Gockenbach's book is only a guide line. Several books in the recommended texts are very good sources. Each book has its advantages and drawbacks. I will provide my lecture notes for this class.

Week 1: (Gockenbach §1, §2 & part of §3)

Linear system, from set to field, Gaussian elimination.

Matrix algebra.

Vector spaces over fields, subspaces.

Week 2: (Gockenbach §3)

Linear transformation and their matrices,

Independence, basis and dimension.

Change of coordinates

Week 3 (Gockenbach §4)

Determinants. Diagonalization.

Eigenvalues and eigenspaces.

Week 4: (Gockenbach §5)

Jordan canonical forms.

Discrete Dynamical System; Perron-Frobenius Theorem; Markov chain

Week 5: (Gockenbach §6 §7)

Inner Products, Projections, Orthogonal basis, Gram-Schmidt process, QR factorization,

Least-squares approximation. FFT

Week 6: (Gockenbach §8)

The spectral theory of symmetric matrices.

Quadratic forms, positive and positive-definite transformations and matrices.

The singular value decomposition and applications.

Weeks 7

Review / extra topics / Presentation

Computer lab topics:

Solving Linear Equations

Recover matrix from data

Change of coordinates for circular orbit

Area and determinant

Discrete Dynamical Systems

Least Squares Data Fitting

FFT

Digital Image Compression (SVD)

etc.